variants registered in the COSE algorithms registry (https://www.iana.org/assignments/cose/cose.xhtml#algorithms), the vast majority of which are marked as recommended. This desire to cover all bases can make it hard for developers to know which algorithms to choose and while many of them are fine algorithms, they can lead to security issues when misused, such as the accidental nonce reuse issues you've learned about in the last few sections.

SHA-3 and STROBE

The US National Institute of Standards and Technology (NIST) recently completed an international competition to select the algorithm to become SHA-3, the successor to the widely used SHA-2 hash function family. To protect against possible future weaknesses in SHA-2, the winning algorithm (originally known as Keccak) was chosen partly because it is very different in structure to SHA-2. SHA-3 is based on an elegant and flexible cryptographic primitive known as a *sponge construction*. Although SHA-3 is relatively slow in software, it is well suited to efficient hardware implementations. The Keccak team have subsequently implemented a wide variety of cryptographic primitives based on the same core sponge construction: other hash functions, MACs, and authenticated encryption algorithms. See https://keccak.team for more details.

Mike Hamburg's STROBE framework (https://strobe.sourceforge.io) builds on top of the SHA-3 work to create a framework for cryptographic protocols for IoT applications. The design allows a single small core of code to provide a wide variety of cryptographic protections, making a compelling alternative to AES for constrained devices. If hardware support for the Keccak core functions becomes widely available, then frameworks like STROBE may become very attractive.

If you need standards-based interoperability with other software, the COSE can be a fine choice for an IoT ecosystem, so long as you approach it with care. In many cases, however, interoperability is not a requirement because you control all of the software and devices being deployed. In this a simpler approach can be adopted, such